CASE REPORT WITH MINI-REVIEW

Recurrent Mycoplasma genitalium Urethritis with Sequential Treatment Failures:

Diagnostic Gaps, Macrolide Resistance, and Fluoroquinolone-Mineral Chelation as Compounding Barriers to Cure

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ABSTRACT

Background. *Mycoplasma genitalium* (M. gen) is an established cause of nongonococcal urethritis (NGU) with rapidly increasing antimicrobial resistance. Resistance-guided sequential doxycycline-fluoroquinolone therapy is now the standard of care in many settings, but treatment failure continues to occur even with apparently appropriate regimens. Pharmacokinetic interactions between fluoroquinolones and dietary mineral supplements represent an underrecognized mechanism of therapeutic failure.

Case Summary. We report a 34-year-old immunocompetent male with recurrent *M. genitalium* urethritis over a four-month period following urethral instrumentation. Despite multiple antibiotic courses including azithromycin, moxifloxacin, and prolonged sequential doxycycline-moxifloxacin therapy, NAAT positivity and symptoms persisted. Microbiological cure was achieved only after the patient self-discontinued a multivitamin containing iron, zinc, calcium, and magnesium that he had been co-administering with moxifloxacin throughout each treatment course.

Conclusion. This case illustrates three compounding barriers to cure in *M. genitalium* infection: (1) empiric azithromycin without resistance testing driving macrolide resistance via the single-copy 23S rRNA architecture; (2) pharmacokinetic failure from fluoroquinolone-mineral chelation reducing moxifloxacin bioavailability; and (3) prolonged antibiotic exposure resulting in recurrent leukopenia. Clinicians should routinely inquire about dietary supplement use when managing M. gen treatment failure, and resistance-guided therapy with meticulous medication counseling should be standard.

Keywords: *Mycoplasma genitalium*; urethritis; macrolide resistance; moxifloxacin; fluoroquinolone chelation; pharmacokinetics; treatment failure; antimicrobial stewardship

1. Introduction

Mycoplasma genitalium was first isolated from men with nongonococcal urethritis in 1981 and is now recognized as the second most common bacterial cause of NGU after *Chlamydia trachomatis*, accounting for 20–30% of male urethritis cases in clinical settings.[1] The organism is fastidious, unculturable by routine methods, and detectable only by nucleic acid amplification testing (NAAT), a circumstance that delayed recognition of its clinical importance by decades.[2]

The emergence of antimicrobial resistance has dramatically complicated management. Macrolide resistance-associated mutations in the 23S rRNA gene rose from 10% globally before 2010 to 51% by 2016–2017, with rates as high as 64% in US sexual health clinics.[3] Fluoroquinolone resistance, mediated principally by parC mutations (S83I being most treatment-predictive), is increasing in parallel, with a global prevalence of 4.3% but substantially higher in some regions.[3] These trends have prompted a paradigm shift from empiric syndromic treatment toward resistance-guided sequential therapy in guidelines from the United States, Europe, and Australia.[4−6]

Despite this evolution, treatment failure persists even in patients receiving apparently guideline-concordant therapy. Pharmacokinetic (PK) barriers represent an underappreciated cause of failure. Fluoroquinolones form insoluble chelation complexes with divalent and trivalent cations—including iron, zinc, calcium, and magnesium—resulting in substantially reduced gastrointestinal absorption.[7] Widely available multivitamin supplements contain all four of these cations and are frequently co-administered with antibiotics without clinician awareness.

We present a case of recurrent *M. genitalium* urethritis with four treatment failures attributable to three distinct mechanisms, culminating in microbiological cure only after identification of a previously unrecognized pharmacokinetic interaction. A mini-review of the relevant epidemiology, resistance biology, pharmacokinetics, and treatment framework is provided.

2. Case Report

2.1 Patient and Presentation

A 34-year-old male with no significant past medical history (incidentally elevated fasting glucose on prior screening) presented to a public health STI clinic in early October 2025 with urethral discharge and dysuria following the onset of a new female sexual partnership and urethral instrumentation (self-

insertion of a silicone catheter as a sexual practice, so-called "sounding"). He was a never-smoker, reported occasional alcohol use, and daily marijuana use.

2.2 Clinical Chronology

The complete clinical timeline is summarized in Table 1. Key events are as follows:

Table 1. Clinical Chronology

Date	Event / Finding	Intervention
Oct 2025 (early)	New female partner; urethral sounding (silicone catheter) as sexual practice	—
Oct 2025	Urethral discharge, dysuria. Health dept: M. gen NAAT+; partner treated	Doxycycline 100 mg BID × 7d
Oct 15, 2025	Asymptomatic. HIV, HCV, syphilis negative	Observation
Oct 21, 2025	Recurrent symptoms. M. gen NAAT+; GC/CT/TV negative	Doxycycline + Azithromycin 1g
Oct 22, 2025	Macrolide resistance presumed; doxy 'debulking' then moxifloxacin planned	Moxifloxacin 400 mg × 7d (post-doxy)
Nov 11, 2025	Symptoms recurred 3–4 days post-moxifloxacin. M. gen NAAT+	Doxycycline 10d then azithromycin 4-day
Nov 29, 2025	Recurrence ~2 days post-antibiotics. Referred to ID and Urology	ID + Urology consultations initiated
Dec 4, 2025	ID consult. Plan: doxycycline + moxifloxacin × 21 days sequential	Combination therapy initiated
Dec 10, 2025	CBC: WBC 3.23 × 10^3/µL; low ANC — leukopenia	Monitoring; therapy continued
Dec 26, 2025	Urology: cystourethroscopy — no stricture, no lesions, no scarring. PVR 6 mL. M. gen NAAT+	No urological intervention required
Jan 5, 2026	UA: 26–50 WBCs/hpf — active urethritis confirmed. Doxycycline restarted	Doxycycline restarted Jan 6
Jan 15, 2026	ID f/u: sequential doxy 14d → moxifloxacin 400 mg × 14d. Resistance genotyping ordered. Scrotal US ordered	Sequential therapy; resistance testing
Jan 15, 2026	Scrotal US: simple 3 mm left epididymal cyst; small bilateral hydroceles — no epididymitis	No intervention
Jan 24, 2026	M. gen NAAT+. Leukocyte esterase negative. WBC 3.12 × 10^3/µL (2nd leukopenia)	Monitoring; therapy ongoing
Jan 28, 2026	Patient self-discontinued multivitamin (taken since fall 2025)	Chelation source removed
Jan 29, 2026	Symptoms resolved (1 day after multivitamin discontinuation)	Therapy continued without supplementation
Feb 3, 2026	Sequential antibiotic course completed (total doxy 14d + moxi 14d)	Awaited TOC
Feb 5, 2026	ID f/u: fluoroquinolone–mineral chelation documented as probable PK failure mechanism	Counselling; no further supplementation
Feb 9, 2026	M. gen NAAT negative — premature (only 6 days	Repeat TOC scheduled

Date	Event / Finding	Intervention
	post-moxifloxacin; below ≥3-week guideline threshold)	
Feb 27, 2026	M. gen NAAT negative — confirmatory TOC (≥3 weeks post-treatment)	Cure documented

M. gen = Mycoplasma genitalium; NAAT = nucleic acid amplification test; GC = Neisseria gonorrhoeae; CT = Chlamydia trachomatis; TV = Trichomonas vaginalis; UA = urinalysis; WBC = white blood cell count; PVR = post-void residual; TOC = test of cure; ID = infectious disease; f/u = follow-up.

2.3 Pivotal Finding: Fluoroquinolone–Mineral Chelation

Throughout the entire treatment course from October 2025 through February 2026, the patient was taking a daily multivitamin supplement containing iron 18 mg, zinc 11 mg, calcium 200 mg, and magnesium 100 mg. Each moxifloxacin course was taken concurrently with this supplement. On January 28, 2026, the patient independently discontinued the multivitamin; symptoms resolved within 24 hours. Microbiological cure was documented on confirmatory TOC on February 27, 2026 (≥3 weeks post-final moxifloxacin dose, consistent with European guideline requirements).[5]

2.4 Hematologic Monitoring

Two episodes of leukopenia were documented during prolonged antibiotic exposure: WBC 3.23 × 10^3/µL with low absolute neutrophil count on December 10, 2025, and WBC 3.12 × 10^3/µL on January 24, 2026. Both resolved without specific intervention after antibiotic course completion. Leukopenia is a recognized but incompletely characterized adverse effect of prolonged doxycycline and moxifloxacin therapy; routine CBC monitoring during extended courses is warranted.

3. Discussion and Mini-Review

3.1 Epidemiology and Natural History

3.1.1 General Population Prevalence

M. genitalium is among the most prevalent sexually transmitted bacterial infections in high-income countries. The most comprehensive meta-analysis to date (Baumann et al. 2018, 9 studies, n > 50,000) estimated general population prevalence at 1.3% (95% CI 1.0–1.7%) in high-HDI countries, with prevalence among asymptomatic STI clinic attendees of 0.8% (I^2 = 0%).[8] These figures place *M. genitalium* in the same order of magnitude as *Chlamydia trachomatis* (US general population prevalence 1.7–2.0%), with which it shares clinical presentation, tissue tropism, and a propensity for asymptomatic carriage. Population-based cohort data from Australia (Machalek et al. 2020)

demonstrate prevalence of 1.0–2.0% in men and 1.0–3.3% in reproductive-age women, with peak prevalence in the 18–34 age group.[7] Applying a conservative 1.3% point estimate to the US adult population of approximately 258 million implies roughly 3.3 million prevalent infections at any given time — a burden comparable in scale to gonorrhea, and almost entirely unsurveilled.

3.1.2 Prevalence in High-Risk and Clinical Populations

Prevalence rises substantially in symptomatic and at-risk populations. Among men with nongonococcal urethritis (NGU), *M. genitalium* is identified in 15–25% of cases; in recurrent or persistent NGU, the proportion increases to 20–40%, reflecting both the organism's tendency for treatment failure and the clinical selection bias of repeated presentations. In MSM attending STI clinics, prevalence ranges from 7–15%. Among women with cervicitis, *M. genitalium* is identified in 10–30% of cases; in women with PID, approximately 10–15%. HIV-positive men carry rates as high as 29% in some cohort studies. These figures collectively establish *M. genitalium* as a clinically significant pathogen across all major at-risk groups, yet one that remains largely absent from routine diagnostic panels in the United States.

3.1.3 Co-infection with Chlamydia trachomatis: Rates and Stewardship Implications

Co-infection of *M. genitalium* with *Chlamydia trachomatis* is common, clinically underrecognized, and has direct antimicrobial stewardship consequences. Co-infection rates are highly population-dependent, ranging from 3% in general STI clinic cohorts to 36–38% in high-risk clinic populations specifically selected for CT infection. In a large US multicenter clinical cohort, *M. genitalium*/CT dual infection was present in 3.1% of all female STI patients, with the majority of *M. genitalium* infections (68.7% in women, 71.6% in men) presenting as single-organism infections.34 Among young high-risk women with asymptomatic bacterial vaginosis, *M. genitalium* co-infected 24–30% of women who had either chlamydial or gonococcal infections.35 In a Birmingham STI clinic cohort specifically studying CT-infected women, *M. genitalium* co-infection was present in 7.3%, though prior US clinic-based studies found rates of 25–36%.36

The stewardship implication is direct and serious. Azithromycin 1g — the historical standard treatment for chlamydia — is simultaneously empiric macrolide monotherapy for any undiagnosed co-existing *M. genitalium* infection. In the Birmingham cohort, among co-infected women who received azithromycin 1g for CT treatment, 28.6% had *M. genitalium* detected again at 3-month follow-up — and all persistent infections harbored macrolide resistance-associated mutations in the

23S rRNA gene.36 Every patient treated for chlamydia without *M. genitalium* co-testing is therefore a potential resistance selection event: the azithromycin cures the chlamydia and simultaneously selects for macrolide-resistant *M. genitalium* in the co-infected patient who departs the clinic never diagnosed. This mechanism is directly relevant to the present case: the October 21, 2025 encounter added azithromycin 1g to doxycycline in the context of a documented *M. genitalium* NAAT positive — precisely the treatment scenario that drives macrolide resistance selection. The transition to resistance-guided therapy and the eventual shift away from empiric azithromycin in CT management are among the most important near-term stewardship interventions available for both infections simultaneously.

3.1.4 Surveillance Gaps and the US Context

Unlike gonorrhea, chlamydia, and syphilis, *M. genitalium* is not a reportable condition in any US state, and the CDC does not include it in national STI surveillance reporting. No nationally representative US prevalence study equivalent to the UK's National Survey of Sexual Attitudes and Lifestyles (Natsal) exists. This represents a fundamental epidemiological blind spot at a moment when resistance rates are rising rapidly. Several factors compound underestimation: approximately 50% of infected men and up to 70% of infected women are asymptomatic and never seek care; the organism cannot be cultured in standard clinical laboratories, requiring explicit NAAT order; the ResistancePlus MG resistance testing platform (SpeeDx) is not FDA-approved, limiting the utility of testing in settings where the result cannot guide optimal treatment; and many clinicians remain unaware of the organism's clinical significance or the availability of diagnostic testing.

The disparity between the public health infrastructure for *M. genitalium* and that for *Chlamydia trachomatis* — an organism it closely resembles in nearly every clinically relevant dimension — is difficult to justify on epidemiological grounds. Both organisms affect approximately 1.3–2.0% of the general population; both are asymptomatic in roughly 50% of infected men and 70–80% of infected women; both colonize the same urogenital epithelial surfaces via similar mechanisms; and both cause identical upstream sequelae — cervicitis, endometritis, PID, tubal factor infertility, and ectopic pregnancy — through the same mechanism of subclinical ascending infection. Chlamydia has USPSTF grade B screening recommendations for sexually active women aged ≤24, mandatory reporting in all 50 states, inclusion in standard STI panels, and a well-resourced CDC annual surveillance report. *M. genitalium* has none of these. The critical asymmetry is resistance: *C. trachomatis* has remained universally susceptible to doxycycline after decades of use. *M. genitalium*

macrolide resistance in the US has reached ≥64% and is rising, while fluoroquinolone resistance is emerging globally. The organism that is harder to treat, equally prevalent, equally silent, and equally capable of causing permanent reproductive harm is the one for which the US public health response remains essentially absent. The chlamydia surveillance model took a generation of epidemiological investment to build; the case for applying a similar framework to *M. genitalium* is now compelling.

3.1.5 Temporal Trends and the "Next Gonorrhea" Trajectory

General population prevalence of *M. genitalium* appears stable or slowly increasing, but resistance rates are rising steeply — a dissociation that mirrors the trajectory of *Neisseria gonorrhoeae* through the 1980s and 1990s. US macrolide resistance rates have climbed from approximately 10% in the early 2010s to ≥64% in recent surveillance data; fluoroquinolone resistance in some global populations now exceeds 10%.[3] The convergence of high organism prevalence, high asymptomatic carriage, escalating resistance, absence of national surveillance, and a shrinking therapeutic armamentarium has led prominent investigators to characterize *M. genitalium* as "the next gonorrhea" — an organism on a resistance trajectory that, without systematic intervention, will render current treatment guidelines obsolete within a decade. The present case, in which five sequential antibiotic courses across three drug classes failed before pharmacokinetic failure was identified as the proximate cause, illustrates at the individual patient level the consequences of this trajectory playing out in clinical practice.

3.1.6 Natural History of Untreated Infection

Modeling data from cohort studies (Smieszek and White 2016) suggest a mean untreated infection duration of approximately 15 months, corresponding to a spontaneous clearance rate of $\gamma = 0.80$/year when corrected for study design artifacts and reinfection dynamics.[9] This prolonged natural history carries two compounding implications. First, it sustains an extended transmission window during which an inadequately treated patient remains infectious to partners. Second, and most relevant to the present case, it creates an extended window during which subtherapeutic antibiotic exposure — whether from resistance, pharmacokinetic failure, or both — exerts ongoing selection pressure without achieving microbiological cure. Each course of moxifloxacin administered to this patient while chelation was rendering drug levels subtherapeutic was, in effect, providing a prolonged resistance-selection stimulus to a bacterial population with an already-elevated mutation rate and a

single-copy rRNA gene architecture permitting immediate, complete macrolide resistance from a single nucleotide substitution.

3.2 Microbiological Characteristics and Resistance Architecture

3.2.1 Genomic Vulnerability to Resistance

M. genitalium harbors a single copy of the 23S rRNA gene, in stark contrast to the 4–7 copies found in most bacteria.[10] This architectural peculiarity has profound pharmacological consequences: in organisms with multiple rRNA operons, a resistance-conferring point mutation initially affects only a fraction of ribosomes, producing a heteroresistant phenotype that may be overcome by higher antibiotic concentrations. In *M. genitalium*, a single nucleotide substitution (most commonly A2058G or A2059G) immediately renders 100% of ribosomes resistant. Azithromycin monotherapy thus functions not merely as inadequate treatment but as a potent selection engine, amplifying pre-existing or treatment-induced resistant clones irreversibly.

Further compounding this vulnerability, *M. genitalium* lacks a functional methyl-directed mismatch repair (MMR) system, resulting in an elevated baseline mutation rate that accelerates the emergence of de novo resistance mutations.[11] Unlike gonorrhea, which acquires resistance through horizontal gene transfer, all *M. genitalium* resistance is chromosomal — once selected, it cannot be lost through plasmid curing or transformation.

3.2.2 Fluoroquinolone Resistance Mechanisms

Fluoroquinolone resistance in M. gen arises through sequential mutations in parC followed by gyrA. The parC G248T (S83I) mutation is both the most prevalent and the most predictive of moxifloxacin treatment failure, present in 4.3% of isolates globally but substantially higher in certain geographic clusters.[3] Concurrent gyrA mutations (M95I, D99N) appear to act synergistically with parC mutations, producing higher-level fluoroquinolone resistance.[12]

In the present case, each moxifloxacin course administered without the recognition of chelation-induced PK failure created precisely the subtherapeutic-level exposure that is most conducive to stepwise parC → gyrA mutation accumulation. Resistance genotyping was ordered but a result was not available prior to documentation of cure.

3.3 Mycoplasma genitalium as an Experimental Model for Resistance Biology

3.3.1 The Minimal Genome Advantage

M. genitalium harbors the smallest known self-replicating bacterial genome — approximately 470 protein-coding genes across 580 kbp. This is not merely a biological curiosity. Because every gene is essential or near-essential, the organism offers a uniquely clean experimental system: there are no redundant paralogues, minimal metabolic buffering, and no confounding background complexity. Transposon mutagenesis screens mapped the complete essential gene set, defining which genes can be disrupted without loss of viability — directly identifying potential drug targets.19 This work established that all 8 ribosomal proteins, FtsZ (cell division GTPase), the Sec translocon, and the complete DNA replication apparatus are essential with no functional redundancy.

The JCVI-syn3.0 synthetic cell (Hutchison et al., *Science* 2016) distilled the genome further to 473 genes — of which approximately 149 have no known function — creating a platform to systematically probe gene essentiality.20 This synthetic chassis allows researchers to introduce foreign genetic circuits, including resistance determinants from other pathogens, and study their behavior in a defined, minimal cellular context. The result is an organism that functions as a living pharmacodynamic test bench: resistance genes can be studied in near-complete isolation from confounding regulatory networks.

3.3.2 The Mutator Phenotype as a Resistance Model

M. genitalium lacks functional homologs of MutL and MutH — the central mismatch repair (MMR) proteins — resulting in a spontaneous mutation rate of approximately 10^{-5} to 10^{-4} per gene per generation, roughly 10- to 100-fold higher than *Escherichia coli*. This constitutive mutator phenotype creates a model directly relevant to clinically important resistance dynamics observed across multiple pathogens:

- Pseudomonas aeruginosa in cystic fibrosis airways — prolonged bronchiectatic colonization selects for MutS-deficient hypermutator clones that rapidly accumulate multidrug resistance. This is one of the best-characterized clinical examples of mutator-driven resistance escalation, and M. gen provides a tractable in vitro surrogate for studying the kinetics of this process.
- Staphylococcus aureus — certain MRSA lineages with partial MMR defects accumulate resistance mutations at accelerated rates under antibiotic selective pressure, particularly during prolonged bacteremia or device-associated infection.
- Mycobacterium tuberculosis — while MTB has a distinct repair architecture, isolates with katG and oxyR loss develop elevated mutation rates; the M. gen mutator model informs thinking about how repair deficiency interacts with prolonged drug exposure in slowly-dividing pathogens.

Because the complete *M. genitalium* repair apparatus is known and small enough to manipulate precisely, defined mutator strains with specific repair gene knockouts can be constructed and their

resistance acquisition kinetics tracked in real time — a controlled experiment that is technically far more demanding in organisms with complex, redundant repair networks.

3.3.3 Serial Passage Resistance Studies and the Single-Copy rRNA Paradigm

M. genitalium can be serially passaged in cell-free SP-4 broth, making it amenable to in vitro resistance selection experiments despite its fastidious growth characteristics. These experiments have generated several findings with broad translational importance:

Stepwise passage in increasing azithromycin concentrations reliably generates A2058G or A2059G point mutations in the single-copy 23S rRNA gene, producing full macrolide resistance within 1–3 passages — far faster than organisms with 4–7 rRNA operons.21 This experimentally validated the clinical observation that a single azithromycin course can drive complete resistance in *M. genitalium*. Parallel experiments with *Helicobacter pylori* (which has 2 copies) and *Mycobacterium tuberculosis* (1 copy, in the 23S context of the ribosomal large subunit) have reinforced the principle: rRNA operon copy number is a primary determinant of how rapidly macrolide resistance can be selected. The single-copy architecture of *M. genitalium* places it at the extreme end of this spectrum — the most resistance-prone configuration biologically possible.

Serial fluoroquinolone passage experiments in *M. genitalium* reproducibly generate the parC → gyrA stepwise mutation pattern now observed clinically, validating the sequential resistance model and demonstrating that the fitness cost of combined parC + gyrA mutations, while greater than parC alone, remains compatible with sustained replication and clinical persistence. This fitness cost framework — established in *M. genitalium* — informs the design of combination regimens intended to impose prohibitive fitness costs on doubly resistant mutants.

3.3.4 Drug Target Discovery

The minimal proteome of *M. genitalium* has made it a productive target discovery platform for novel antibacterial agents. Key contributions include:

- FtsZ inhibitors — FtsZ (the prokaryotic tubulin homolog driving bacterial cytokinesis) is present and essential in M. gen but absent in mammals. Structural and biochemical characterization of M. gen FtsZ contributed to pre-clinical rationale for FtsZ inhibitor programs (including the PC190723 series), which are under development as anti-MRSA agents.
- CARDS toxin — the Community-Acquired Respiratory Distress Syndrome toxin, a unique M. gen ADP-ribosylating and vacuolating virulence factor with mechanistic homology to pertussis toxin, was characterized in this organism. Its catalytic domain represents a potential antibacterial target class whose biology has been clarified using the M. gen minimal-proteome context.

- Ribosome-targeted agents — the complete ribosome structure of M. gen, with all 8 ribosomal proteins essential and no paralogue redundancy, provides a clean structural context for studying novel ribosome-binding antibiotics beyond macrolides, including oxazolidinones, pleuromutilins, and next-generation aminoglycosides.

3.3.5 Translational Relevance of This Case

The pharmacokinetic failure documented in this case has direct experimental relevance: the chelation-induced subtherapeutic moxifloxacin exposure (estimated AUC reduction ~39%, Cmax reduction ~59%) represents a defined, quantifiable form of inadequate drug exposure. In a serial passage model, exposing *M. genitalium* to moxifloxacin concentrations mimicking the chelated pharmacokinetic profile — approximately 61% of therapeutic AUC — would be expected to preferentially select for parC mutations while failing to achieve bacterial clearance. This experiment would directly model the clinical course of this patient: four courses of inadvertently subtherapeutic moxifloxacin, each potentially advancing the organism along the parC → gyrA resistance trajectory without achieving cure. Such a study would constitute a controlled pharmacokinetic-resistance acquisition model, and the present case report provides its clinical rationale.

3.4 Pharmacokinetic Failure: The Fluoroquinolone–Mineral Chelation Mechanism

Fluoroquinolones chelate divalent and trivalent metal cations (Fe^{2+}, Zn^{2+}, Ca^{2+}, Mg^{2+}, Al^{3+}) through coordination bonds involving the 3-carboxyl and 4-keto groups of the quinolone nucleus.[7] The resulting complexes are insoluble and non-absorbable from the gastrointestinal tract, effectively sequestering the antibiotic before systemic absorption can occur. For moxifloxacin specifically, co-administration with ferrous sulfate reduces the area under the curve (AUC) by up to 39% and peak concentration (Cmax) by up to 59% in pharmacokinetic studies.[13]

The patient's daily multivitamin contained: iron 18 mg (as ferrous fumarate), zinc 11 mg, calcium 200 mg, and magnesium 100 mg. Under standard dosing, moxifloxacin achieves an oral bioavailability of ~86% with a half-life of approximately 12 hours; peak serum concentrations of 3.1–3.9 µg/mL are reached within 0.5–4 hours.[13] With chelation reducing AUC by 39% and Cmax by ~59%, effective serum levels likely remained below the pharmacodynamic target (AUC/MIC ratio required for bacterial killing), converting each moxifloxacin course into a de facto subtherapeutic exposure for the full duration of supplement co-administration.

The clinical significance of the chelation event is supported by the timeline: the patient had taken moxifloxacin four times without microbiological cure while co-administering the multivitamin. The final course, comprising doxycycline 100 mg BID × 14 days followed by moxifloxacin 400 mg daily × 14 days, achieved microbiological cure only after supplement discontinuation on day 28 of the overall course — within the moxifloxacin treatment window. No other clinical variables changed.

This interaction is not idiosyncratic: all currently available fluoroquinolones are susceptible to cation chelation to varying degrees, and multivitamin supplement use is common and rarely elicited in antibiotic history-taking. The 2021 CDC STI Guidelines and standard pharmacokinetic references note this interaction for fluoroquinolones as a class, but clinical practice patterns suggest it is underappreciated in the context of *M. genitalium* management specifically.

3.5 The "Debulking" Strategy: Bacterial Load Reduction Before Definitive Therapy

3.5.1 Rationale in Mycoplasma genitalium

The term "debulking" — borrowed from oncological surgery — refers in antimicrobial therapeutics to the intentional reduction of pathogen burden as a preparatory step before definitive eradication therapy. In the context of *M. genitalium*, the 2021 CDC STI Guidelines recommend initiating therapy with doxycycline 100 mg BID × 7 days before moxifloxacin, explicitly acknowledging that doxycycline achieves microbiological cure in only 30–45% of cases.4 The intended effect of this step is not cure but load reduction: a smaller pathogen population entering the moxifloxacin phase means fewer organisms carrying pre-existing fluoroquinolone resistance mutations, and lower probability that de novo mutations emerge during treatment.

The mathematical basis for this strategy derives from Luria-Delbruck mutation dynamics. In any sufficiently large clonal population, the probability that at least one organism harbors a pre-existing resistance mutation approaches certainty as population size increases. Conversely, reducing the inoculum by even one order of magnitude substantially decreases the expected number of resistant mutants and the probability that selection and clonal expansion occur.15 For *M. genitalium*, which cannot acquire resistance through horizontal gene transfer and depends entirely on de novo chromosomal mutation, inoculum reduction before definitive therapy is pharmacodynamically rational.

An important unresolved tension, explicitly noted by Manhart and colleagues, is that the mechanism underlying doxycycline's partial efficacy in *M. genitalium* infection is unknown.11 No tetracycline resistance mechanisms have been characterized in this organism, suggesting either partial bacteriostatic activity that facilitates immune clearance, host-mediated pathways, or uncharacterized intrinsic resistance determinants in the non-cured fraction. The clinical adoption of doxycycline as a load-reducing step predates mechanistic confirmation of this function.

3.5.2 The Inoculum Effect: A Parallel Pharmacodynamic Consideration

The inoculum effect describes the phenomenon whereby antibiotic minimum inhibitory concentrations increase in proportion to bacterial density — most classically with beta-lactams and beta-lactamase-producing Staphylococcus aureus, but applicable more broadly. At high bacterial density, enzymatic drug inactivation or efflux can outpace antibiotic-mediated killing, rendering a drug functionally ineffective at concentrations that would otherwise be bactericidal. Load reduction strategies directly counter the inoculum effect by shifting the pharmacodynamic balance in favor of the antibiotic before definitive therapy begins. This concept is directly relevant to high-burden STI presentations and partially explains why cure rates for guideline-concordant moxifloxacin therapy are higher when preceded by doxycycline debulking than when moxifloxacin is used as monotherapy.

3.5.3 Cross-Pathogen Parallels — Table 2

The debulking concept has established or proposed application across several resistant pathogens (Table 2). Key comparators are discussed below.

Table 2. Debulking Strategies Across Clinically Relevant Resistant Pathogens

Organism / Condition	Debulking Agent(s)	Definitive / Consolidation Agent	Evidence Level
Cryptococcus neoformans meningitis	AmB + flucytosine × 2 wks (induction)	Fluconazole 400–800 mg/d × 8 wks, then 200 mg/d maintenance	Strong — RCT (ACTG 5164, ACTA trial)16
Mycobacterium tuberculosis (drug-susceptible)	HRZE × 2 months — intensive phase cavity sterilization	HR × 4 months — continuation phase	Strong — foundational to WHO framework
Helicobacter pylori (clarithromycin-resistant settings)	Bismuth quadruple therapy (PPI + bismuth + tetracycline + metronidazole)	Rifabutin-based or vonoprazan-amoxicillin rescue	Moderate — resistance prevalence >15% drives bismuth-first strategy17
MRSA endovascular (high inoculum)	Daptomycin + beta-lactam (synergistic membrane-wall stress at high density)	Definitive daptomycin or alternative per susceptibility	Moderate — observational; IDSA guidelines18

Organism / Condition	Debulking Agent(s)	Definitive / Consolidation Agent	Evidence Level
Mycoplasma genitalium	Doxycycline 100 mg BID × 7–14d (30–45% cure; load-reduction function)	Moxifloxacin 400 mg × 14d	Moderate — 2021 CDC; mechanism not fully defined4

AmB = amphotericin B; HRZE = isoniazid + rifampin + pyrazinamide + ethambutol; HR = isoniazid + rifampin; PPI = proton pump inhibitor; MRSA = methicillin-resistant Staphylococcus aureus.

3.5.4 The Strongest Analogy: Cryptococcal Meningitis

Cryptococcal meningitis represents the most rigorously studied application of antimicrobial debulking. *Cryptococcus neoformans* achieves CSF burdens exceeding 10^5 CFU/mL in untreated infection — a density at which fluconazole, despite excellent CNS penetration, is fungistatic and cannot achieve the rapid fungal clearance necessary to prevent IRIS or early mortality. Induction with amphotericin B deoxycholate plus flucytosine (flucytosine providing complementary inhibition of fungal RNA synthesis) for 2 weeks achieves greater than a 10-fold reduction in CSF fungal burden before fluconazole consolidation begins. The ACTA trial demonstrated that this induction combination reduced 10-week mortality from 41% (fluconazole alone) to 24%, with the greatest benefit attributable to the rate of early fungal clearance.16 The parallels with *M. genitalium* are direct: (1) initial agent reduces burden without necessarily achieving cure; (2) definitive agent is deployed against a reduced, less resistance-enriched population; (3) each agent acts via a distinct mechanism, limiting cross-resistance.

3.5.5 Tuberculosis: The Canonical Model

The two-phase structure of tuberculosis therapy — an intensive phase (HRZE × 2 months) followed by a continuation phase (HR × 4 months) — is the canonical model of antimicrobial debulking in clinical practice. The intensive phase targets all three metabolically distinct subpopulations of *M. tuberculosis* (rapidly dividing bacilli, semi-dormant persisters in acidic environments, and intermittently dividing bacilli in closed lesions) to achieve steep early bactericidal activity and cavity sterilization. The continuation phase then eliminates the residual persister population, which — having been depleted of its most numerous and rapidly dividing members by the intensive phase — carries proportionally lower resistance-mutation probability. This principle directly explains why shortening the intensive phase increases relapse rates: the continuation phase faces a higher residual inoculum, and the probability of acquired drug resistance scales accordingly.

3.5.6 Limitations of Debulking in M. genitalium

The debulking strategy in *M. genitalium* carries important caveats. First, if doxycycline acts as merely bacteriostatic in the non-cured fraction, it may suppress susceptible organisms while resistant subclones continue replicating — potentially enriching, rather than depleting, the resistance-bearing population. Second, without standardized quantitative NAAT or culture methods for *M. genitalium*, actual bacterial load reduction by doxycycline cannot be directly measured in vivo, leaving the debulking function inferential. Third — most relevant to this case — debulking provides no benefit if the definitive agent is pharmacokinetically compromised. Four doxycycline-then-moxifloxacin courses failed when chelation prevented moxifloxacin from reaching therapeutic concentrations, illustrating that the pharmacokinetic integrity of the second-phase agent is the irreducible prerequisite for sequential debulking strategies.

3.6 Diagnostic Framework and Resistance Testing

Etiologic diagnosis of *M. genitalium* requires FDA-approved NAAT (Hologic Aptima or Roche Cobas TV/MG). Critically, resistance-guided therapy — which achieves 92–95.4% cure rates — requires a concurrent resistance testing platform. In the United States, no FDA-approved assay for *M. genitalium* macrolide resistance mutation detection exists at the time of this case; the ResistancePlus MG assay (SpeeDx) is approved in Australia, the UK, Europe, and Canada.[4] US clinicians are therefore largely dependent on empiric sequential therapy or laboratory-developed test platforms.

The 2021 CDC guidelines recommend: (1) doxycycline 100 mg BID × 7 days as initial therapy for NGU (acknowledging only 30–45% microbiological cure for *M. genitalium* specifically, but used as a load-reducing step); (2) moxifloxacin 400 mg daily × 7–14 days when resistance testing is unavailable; and (3) extended moxifloxacin courses (14 days) for recurrent/persistent infections.[4] The present case eventually received a 14-day moxifloxacin course — the maximum recommended duration — but cure was contingent on resolving the chelation barrier.

3.7 Treatment of Refractory Disease: Salvage Options

For patients failing standard sequential therapy, the evidence base is limited to small case series and observational data. Available options include:

- Minocycline 100 mg BID × 14 days: 71% cure in an Australian cohort; may be attempted as a non-fluoroquinolone alternative when parC/gyrA resistance is suspected.[3]
- Pristinamycin 1 g TID × 10 days (in combination with doxycycline): ~75% cure; not available in the United States.[3]

- Sitafloxacin (fourth-generation fluoroquinolone with greater potency against parC mutants): higher cure rates than moxifloxacin in parC S83I-containing infections; not available in the US or most Western markets.[12]
- M3P regimen (minocycline + moxifloxacin + pristinamycin): emerging case series data for multidrug-resistant M. gen; extremely limited availability.[14]
- Gepotidacin (triazaacenaphthylene): low in vitro MICs against M. gen, with in vitro synergy with doxycycline; no published human efficacy data for this indication.[3]

3.8 Test of Cure: International Frameworks and the Philosophical Divide

3.8.1 The Biological Rationale for TOC Timing

Test of cure following *M. genitalium* treatment requires careful attention to NAAT kinetics. NAATs detect nucleic acid rather than viability — degrading DNA from successfully killed organisms persists in the urogenital tract for 14–21 days post-treatment and produces positive results that cannot be distinguished from persistent viable infection. Premature TOC therefore carries the risk of false reassurance: a negative result before the DNA clearance threshold is passed does not confirm microbiological cure; a positive result within the same window may represent residual non-viable DNA rather than treatment failure. The minimum post-treatment interval before TOC is biologically determined, and the near-universal consensus across all major guidelines is ≥3 weeks. In the present case, an initial NAAT performed at 6 days post-moxifloxacin (February 9, 2026) was negative; this result was appropriately discounted as premature. The confirmatory TOC at ≥3 weeks (February 27, 2026) established cure.

3.8.2 The Core Philosophical Divide

Beyond the question of timing, there is a deeper international disagreement about the purpose of TOC — and who should receive it. The divide is essentially this: is TOC a clinical management tool for the individual patient, or a resistance surveillance instrument for the population? Different guidelines answer that question differently, and the answer determines who gets tested, when, and why.

3.8.3 United States — CDC 2021: Selective, Resistance-Contingent TOC

The 2021 CDC guidelines recommend TOC only in specific circumstances: when resistance testing was unavailable and moxifloxacin could not be used, or when alternative or investigational regimens were employed.[4] Routine TOC after standard sequential doxycycline-then-moxifloxacin therapy is not recommended in the absence of persistent symptoms. The implicit assumption is that guideline-

concordant therapy with an agent of known efficacy carries a sufficiently high cure probability that TOC does not change management in most cases.

This position is difficult to defend epidemiologically at a moment when US macrolide resistance has reached ≥64% and fluoroquinolone resistance is emerging. "Guideline-concordant therapy" is a moving target in an organism whose resistance rates are rising faster than guidelines are revised. The honest interpretation of the CDC position is that it reflects the diagnostic infrastructure available in the US context — *M. genitalium* NAAT is not in standard panels, ResistancePlus MG is not FDA-approved, and resistance-guided approaches are not universally accessible — rather than a reasoned independent epidemiological stance. A guideline built around the absence of adequate infrastructure is pragmatic but not aspirational.

3.8.4 Europe — IUSTI/WHO European Guideline 2022: Universal TOC, Resistance-Guided

The European IUSTI guideline (Jensen et al. 2022) recommends TOC for all patients treated for *M. genitalium*, at a minimum of ≥3 weeks post-treatment completion.[5] TOC serves three simultaneous purposes in the European framework: individual clinical management (confirming cure and identifying treatment failure before re-exposure of partners occurs); resistance surveillance (NAAT-positive TOC specimens submitted for resistance genotyping generate real-world prevalence data that feed back into guideline revision); and stewardship (identifying macrolide failure triggers resistance-guided escalation rather than empiric re-treatment, conserving moxifloxacin and reducing fluoroquinolone resistance selection pressure). This framework is enabled by the availability of ResistancePlus MG across EU and UK clinical settings, where resistance-guided therapy achieves cure rates of 92–95.4% — an outcome that justifies the investment in universal TOC by closing the loop between diagnosis, treatment, and outcome.

3.8.5 Australia — ASHM: Universal TOC, Most Aggressive Globally

Australia has developed the most fully realized *M. genitalium* management framework in the world, reflecting both the earliest recognition of the resistance crisis and the most mature surveillance infrastructure. Australian guidelines recommend universal TOC at ≥3 weeks for all treated patients, resistance-guided therapy as the standard of care rather than the exception, re-treatment guided by resistance genotyping of TOC-positive specimens, and TOC for female partners of infected men. Australia's sustained investment in universal TOC has generated the most robust longitudinal

resistance surveillance dataset of any country — the resistance trajectories documented in Australian cohorts have driven urgency in international guideline development globally. The philosophy is explicitly populational: every TOC-positive specimen is simultaneously a clinical event and a resistance surveillance data point.

3.8.6 United Kingdom — BASHH: Universal TOC, Integrated into National Infrastructure

The British Association for Sexual Health and HIV (BASHH) guidelines align closely with the European IUSTI framework — universal TOC at ≥3 weeks, resistance-guided therapy as standard. The UK has integrated *M. genitalium* NAAT and resistance testing into its national sexual health clinic network more systematically than any other country, with both the Aptima *M. genitalium* assay and ResistancePlus MG routinely available. TOC for sexual partners and partner notification programs are explicitly incorporated into the BASHH framework, extending the individual clinical event into a network-level intervention.

3.8.7 Canada — PHAC: Selective TOC, Evolving Toward Universal

The Public Health Agency of Canada guidelines occupy an intermediate position: TOC is recommended after treatment for symptomatic *M. genitalium* urethritis and cervicitis at 3–4 weeks post-treatment, but not universally mandated for all treated patients. Canada's position is more aggressive than the current CDC stance but less systematic than the Australian or UK frameworks, reflecting an evolving recognition of the resistance problem within a healthcare system that has not yet fully deployed resistance-guided diagnostic infrastructure at scale.

3.8.8 Comparative Summary and Implications

Guideline	TOC Recommendation	Minimum Interval	Resistance Testing	Philosophy
CDC 2021 (US)	Selective — treatment failure cases only	≥3 weeks (implied)	Not FDA-approved	Infrastructure-limited
IUSTI/WHO 2022 (EU)	Universal	≥3 weeks	Standard	Individual + surveillance
ASHM (Australia)	Universal + partners	≥3 weeks	Standard	Populational surveillance
BASHH (UK)	Universal + partners	≥3 weeks	Standard	Network-level intervention
PHAC (Canada)	Selective-to-universal	3–4 weeks	Evolving	Intermediate

The international divergence on TOC reflects a structural reality: the United States is the only major high-income country managing *M. genitalium* without a functional resistance surveillance

infrastructure, and its selective TOC philosophy is a consequence of that deficit rather than a reasoned independent epidemiological position. The appropriate response is investment in diagnostic infrastructure — FDA approval of a resistance testing platform, inclusion of *M. genitalium* NAAT in standard STI panels, and mandatory reporting — not continued adaptation of clinical guidelines to the limitations of an inadequate system. The chlamydia surveillance model, which took a generation of investment to build, provides the template. The present case illustrates with clinical precision what the consequences of that infrastructure gap look like at the individual patient level: five treatment courses, three drug classes, five months of active infection, and a cure ultimately achieved by recognizing a pharmacokinetic interaction that systematic TOC and resistance genotyping — as practiced in Australia, the UK, and Europe — would have prompted clinicians to investigate far earlier.

3.9 Hematologic Considerations in Prolonged Therapy

Leukopenia and neutropenia during prolonged antibiotic courses are recognized but incompletely characterized phenomena. Doxycycline can suppress bone marrow through interference with protein synthesis in rapidly dividing hematopoietic precursors; moxifloxacin and other fluoroquinolones have been associated with neutropenia and agranulocytosis, particularly with extended exposure. In the present case, two consecutive CBCs demonstrated leukopenia — both occurring during periods of simultaneous doxycycline and moxifloxacin use. Routine CBC monitoring is advisable when antibiotic courses extend beyond 14 days, particularly with combinations involving tetracyclines and fluoroquinolones.

3.10 Future Directions: Targeting Adhesion as a Resistance-Agnostic Strategy

3.10.1 Nutritional Biology and Tissue Tropism

M. genitalium is an obligate nutritional scavenger. Its stripped-down genome encodes no de novo biosynthetic pathways for amino acids, nucleotides, or lipids — all must be acquired from the host. Its primary energy source is glucose, fermented via a truncated glycolytic pathway to lactic acid in the absence of a functional tricarboxylic acid cycle. It scavenges cholesterol and fatty acids directly from host cell membranes, and it incorporates host-derived lipids into its own membrane — a consequence of having no cell wall and therefore no independent membrane synthetic capacity.22

This metabolic dependency explains the organism's tissue tropism with unusual precision. *M. genitalium* colonizes the columnar epithelium of the mid-urethra and endocervix — not the bladder.

The bladder urothelium, while anatomically adjacent, is an inhospitable niche: transitional (umbrella) cells turn over at approximately 100–200 days, providing minimal membrane lipid flux for scavenging; the urothelial surface is protected by a dense glycosaminoglycan (GAG) coat of hyaluronic acid and chondroitin sulfate that physically resists bacterial adhesion; and the hyperosmolar, acidic bladder lumen is nutritionally barren for an organism requiring close epithelial contact to access interstitial substrates. By contrast, urethral columnar epithelium turns over approximately every 2–4 weeks, expresses heparan sulfate proteoglycans (HSPGs) and sialic acid residues densely on its apical surface, and is bathed in subepithelial interstitial fluid rich in glucose and amino acids. The organism's adhesin complex evolved to target precisely this niche — and is essentially non-viable outside it.

3.10.2 The Tip Organelle Adhesin Complex as a Therapeutic Target

Attachment to host epithelium is mediated by a polar tip organelle — a flask-shaped protrusion housing the major adhesin proteins P140 (MgPa) and P110, which form an obligate surface-exposed receptor-binding complex.23 Disruption of either component abolishes adhesion and prevents infection in cell culture models. The host-side receptors are HSPGs and sialic acid residues. Several compounds have been shown to interfere with this interaction experimentally:

- Heparin and heparin-like glycosaminoglycans — competitive inhibition of HSPG binding sites on host epithelium, demonstrated in vitro; systemic use impractical due to anticoagulant effects, but topical urethral formulation theoretically feasible.
- Lactoferrin — the iron-binding glycoprotein present in seminal fluid and mucosal secretions — inhibits M. genitalium adhesion in vitro, possibly through direct interaction with surface adhesin proteins or iron sequestration disrupting surface protein conformation.
- Anti-P140/P110 neutralizing antibodies — block attachment in cell culture; form the conceptual basis for a M. genitalium vaccine, for which the target is experimentally validated though no clinical candidate yet exists.

3.10.3 A Three-Phase Hypothesis: Debulk, Detach, Flush

The nutritional biology of *M. genitalium* suggests a therapeutic framework that is entirely resistance-agnostic — bypassing the macrolide and fluoroquinolone resistance problem by targeting the host-pathogen interface rather than bacterial metabolism or replication. We propose the following three-phase hypothesis for experimental investigation:

Phase 1 — Bacterial load reduction (debulking): Doxycycline 100 mg BID × 7–14 days as currently practiced. Reduces the total organism burden and lowers the probability that the subsequent phase

encounters a large resistant population. As discussed in Section 3.5, the mechanism of doxycycline's partial efficacy remains incompletely characterized, but the load-reduction function is clinically adopted and rational.

Phase 2 — Adhesin blockade: Introduction of a compound targeting either the bacterial tip organelle adhesin complex (P140/P110) or the host HSPG/sialic acid binding sites. This phase does not need to kill the organism — it needs only to prevent anchoring. Organisms unable to adhere to urethral columnar epithelium lose access to the subepithelial nutrient environment they depend on. Critically, detachment from epithelium does not merely displace the organism — it deprives it of its only viable nutrient source, converting the anti-adhesin effect into a de facto starvation mechanism. No anti-adhesin compound specific to *M. genitalium* is currently in clinical development, though the target class is validated and the precedent for anti-adhesin therapy exists in other pathogens: D-mannose competitively inhibits FimH-mediated *E. coli* uropathogen adhesion, and bismuth compounds partially exert their anti-*H. pylori* effect through adhesion disruption.

Phase 3 — Mechanical clearance: Organisms detached from urethral epithelium during Phase 2 enter the urethral lumen and are carried by urine flow into the bladder. The bladder constitutes a physiological kill zone for *M. genitalium*: its dense GAG coat and slowly renewing urothelium (100–200 day turnover) provide no adhesion substrate and no nutrient access; the hyperosmolar acidic urine is hostile to the organism; and normal voiding mechanics provide continuous mechanical clearance. Re-colonization of urethral epithelium from above is therefore unlikely even for organisms that survive brief bladder exposure. Adequate hydration and voiding frequency during this phase would be expected to maximize clearance kinetics.

The conceptual strength of this approach is that resistance mutations in 23S rRNA, parC, or gyrA are entirely irrelevant to its mechanism of action. An organism that is macrolide-resistant, fluoroquinolone-resistant, or both is equally unable to adhere without its tip organelle intact, and equally dependent on epithelial contact for nutrient acquisition. This framework therefore offers a potential path forward for truly refractory multidrug-resistant *M. genitalium* infection that does not depend on the development of new antibiotics.

Key unanswered questions for experimental investigation include: (1) identification of a clinically deliverable anti-adhesin compound with acceptable urethral tolerability; (2) optimal timing and sequencing of the three phases; (3) whether Phase 2 can be achieved with topical rather than systemic

delivery, circumventing the pharmacokinetic challenges that complicated systemic therapy in the present case; and (4) whether this approach is feasible in female patients, where endocervical anatomy presents distinct drug delivery challenges. *M. genitalium*'s tractable in vitro culture system and well-characterized adhesin biology make it an ideal organism in which to test this hypothesis experimentally before clinical translation.

3.10.4 Controlled Epithelial Desquamation: Removing the Substrate

A complementary and largely unexplored strategy extends the anti-adhesin concept by targeting not the adhesin itself but the epithelial cell to which it binds. Rather than blocking attachment, controlled acceleration of urethral epithelial desquamation physically removes the organism's colonization substrate — the surface columnar cell — with the organism attached to it. This approach exploits the stratified architecture of urethral epithelium in a therapeutically favorable way: the surface postmitotic columnar cells that *M. genitalium* colonizes are expendable, already committed to physiological shedding. The regenerative capacity of the urethra resides entirely in the basal cell layer, which is separated from the lumen by 3–5 cell layers, anchored to the basement membrane by high-affinity integrin-mediated adhesion substantially stronger than the desmosomal and glycoprotein-mediated lateral adhesion of surface cells, and inaccessible to luminal chemical agents at concentrations that affect only the surface layer. Urethral columnar epithelium regenerates a complete surface layer within 7–14 days following controlled disruption, as evidenced by post-catheterization recovery.

3.10.5 The Bladder as a Physiological Kill Zone

The anatomical destination of shed cells and detached organisms is critical to understanding why this strategy is not merely displacement. Organisms expelled from the urethral surface enter the bladder — an environment that is actively hostile to *M. genitalium* by every relevant biological parameter. Bladder urothelial umbrella cells turn over at approximately 100–200 days — the slowest epithelial renewal rate in the human body — providing no membrane lipid flux for scavenging. The urothelial surface is protected by a dense glycosaminoglycan coat of hyaluronic acid and chondroitin sulfate that physically resists HSPG-mediated bacterial adhesion. The hyperosmolar, acidic bladder lumen offers no accessible nutrient substrate for an organism requiring close epithelial contact to feed. And normal voiding mechanics provide continuous mechanical clearance. An unanchored *M. genitalium* in the

bladder lumen is therefore not merely displaced — it is delivered into a metabolic dead end from which re-colonization of urethral epithelium is functionally impossible.

A compelling natural precedent validates this concept. Uropathogenic E. coli forms intracellular bacterial communities (IBCs) within bladder epithelial cells, and the bladder's evolved innate defense response is precisely epithelial exfoliation: TLR4 signaling triggers rapid shedding of infected superficial urothelial cells, expelling the intracellular bacterial reservoir into the urine stream. This mechanism — the bladder stripping itself of infected cells — is a conserved innate immune strategy that reduces bacterial burden without antibiotic involvement. M. genitalium appears to evade the analogous urethral response through active suppression of Toll-like receptor signaling via its membrane lipoproteins, which may explain why the organism persists despite a mean untreated duration of approximately 15 months. Facilitating this response exogenously, rather than waiting for it to occur spontaneously, is conceptually sound.

3.10.6 Candidate Pro-Desquamation Approaches

Several agents with established safety profiles in adjacent mucosal contexts merit investigation as pro-desquamation interventions in the urethra:

- Mildly acidified N-acetylcysteine (NAC) irrigant — NAC disrupts disulfide bonds in glycoprotein matrices, loosening surface cell adhesion from the subepithelial scaffold. At pH 4.0–4.5, tolerated by urogenital mucosa (physiological vaginal pH range), it simultaneously creates a chemical environment hostile to M. genitalium while facilitating surface desquamation. NAC has additional direct anti-adhesive properties against several pathogens through HSPG disruption.
- High-volume urethral saline irrigation — mechanical shear from forceful irrigation exceeds the adhesive threshold of chemically destabilized surface cells. Urethral irrigation is standard pre-cystoscopy care and is well-tolerated without permanent epithelial damage. Alone it is likely insufficient to overcome P140/P110-mediated adhesion; in combination with chemical destabilization, the threshold may be achievable.
- Gentle mucosal brushing during cystourethroscopy — a cytology brush passed during cystourethroscopy, identical in principle to cervical Pap smear sampling, can physically dislodge surface columnar cells without basal layer injury. This patient underwent uncomplicated cystourethroscopy; addition of a urethral brush specimen at the time of that procedure would have been straightforward. Urethral epithelium regenerates fully within days to weeks.
- Topical heparin-like glycosaminoglycans — by occupying HSPG binding sites on the epithelial surface before the organism can engage them, these agents simultaneously prevent new attachment and, in the context of chemically loosened surface cells, may reduce the adhesive energy holding already-attached cells in place.

The central unifying principle across all pro-desquamation approaches is that they are entirely resistance-agnostic. A fully multidrug-resistant *M. genitalium* — resistant to macrolides,

fluoroquinolones, and tetracyclines simultaneously — remains dependent on its tip organelle adhesin for epithelial contact, and on epithelial contact for nutrient acquisition. Removing the substrate removes the organism. This framework does not compete with antibiotic therapy; it is designed to operate in sequence with it, and the debulking-detach-flush-desquamate model represents a logical extension of the sequential approach already embedded in current guidelines. Prospective experimental evaluation in *M. genitalium* cell culture and animal models is warranted.

3.11 Host Consequences of Chronic and Undertreated Infection

M. genitalium is not a passive colonizer. Prolonged infection — whether from diagnostic delay, treatment failure, or antimicrobial resistance — imposes a cumulative burden on the host through direct cytotoxicity, sustained mucosal inflammation, immune evasion, and systemic sequelae that extend well beyond the immediate presentation. Understanding these consequences is essential for appreciating the full clinical significance of cases in which adequate treatment is delayed.

3.11.1 Direct Urogenital Pathology in Men

M. genitalium injures urethral epithelium through at least three independent mechanisms. The CARDS toxin (community-acquired respiratory distress syndrome toxin) is an ADP-ribosylating vacuolating toxin that directly damages epithelial cells, inducing cytoplasmic vacuolation, ciliostasis, and cell death independent of the host inflammatory response.25 Simultaneously, membrane lipoproteins activate NF-κB via TLR1/2 signaling, driving sustained local production of pro-inflammatory cytokines including IL-6, IL-8, and TNF-α.26 Hydrogen peroxide produced as a metabolic byproduct contributes additional oxidative epithelial damage. The net result of prolonged infection is chronic mucosal inflammation — documented in the present case by 26–50 WBCs/hpf on urinalysis obtained during active infection. Sustained inflammation at this level carries progressive structural risk: urethral stricture formation, periurethral fibrosis, and luminal scarring. The cystourethroscopy in this case was reassuringly normal; however, that examination was performed at approximately three months into the symptomatic course, not following sustained resolution. The absence of stricture at that point does not exclude development with continued inflammatory burden.

Ascending infection may produce subclinical epididymitis even in the absence of clinically apparent epididymo-orchitis. *M. genitalium* has been recovered from the epididymis and is an established — though underdiagnosed — cause of epididymo-orchitis.27 Of particular importance, *M. genitalium* attaches to spermatozoa via the same P140 tip organelle adhesin used for epithelial colonization,

impairs progressive motility through oxidative membrane damage, and induces sperm DNA fragmentation — effects that can occur without clinically apparent epididymitis.28 Studies in male subfertility cohorts identify *M. genitalium* in 8–10% of men with unexplained reduced fertility, underscoring the reproductive stakes of inadequate treatment.

3.11.2 Reproductive and Obstetric Consequences in Female Partners

The downstream consequences of inadequately treated male infection fall disproportionately on female partners. *M. genitalium* causes cervicitis clinically indistinguishable from chlamydial cervicitis, subclinical endometritis documented by endometrial biopsy, and — most significantly — pelvic inflammatory disease (PID).29 It is the second most common identifiable cause of PID after *N. gonorrhoeae*, found in 10–15% of PID cases. Critically, *M. genitalium*-associated PID is frequently clinically mild ("silent PID"), presenting with minimal or absent systemic features, but causes the same structural tubal damage as symptomatic gonococcal salpingitis: peritubal adhesions, fallopian tube scarring, and the anatomical substrate for tubal factor infertility and ectopic pregnancy.30 Adverse pregnancy outcomes including spontaneous abortion, preterm birth, and premature rupture of membranes have been associated with ascending intrauterine infection, likely through prostaglandin pathway activation. In the present case, the patient's female partner received treatment in October 2025; whether she achieved microbiological cure during the window when the male patient remained infectious — and whether subclinical upper tract involvement developed — is an open question with direct clinical implications.

3.11.3 Immune Evasion and Persistence Mechanisms

M. genitalium sustains chronic infection through several overlapping immune evasion strategies. High-frequency phase variation of MgpB and MgpC surface lipoproteins — mediated by a recombination system generating continuous antigenic novelty — prevents the development of sterilizing antibody responses: the immune system perpetually pursues a shifting antigen repertoire.31 Chronic TLR2/TLR6 activation by membrane lipoproteins ultimately induces endotoxin-like TLR tolerance, progressively anergizing the local mucosal immune response during sustained infection. Most relevant to therapeutic strategy, *M. genitalium* is capable of intracellular invasion of urogenital epithelial cells, where it resides protected from extracellular antibiotics, complement, and antibody-mediated killing.32 This intracellular reservoir represents a plausible mechanism for treatment failures not fully explained by microbiological resistance or pharmacokinetic failure, and provides

additional rationale for the epithelial desquamation strategy described in Section 3.10 — shedding of infected surface cells removes the intracellular reservoir along with the cell.

3.11.4 HIV, HPV, and Oncological Implications

M. genitalium infection increases HIV acquisition risk with an odds ratio of approximately 2–3 through dual mechanisms: mucosal barrier disruption from the inflammatory response enhances HIV transmissibility, and CD4+ T cell recruitment to sites of urogenital mucosal inflammation creates a local enrichment of HIV target cells at the point of exposure.33 HIV shedding in genital secretions is similarly increased in *M. genitalium*-infected individuals, with implications for onward HIV transmission.

The relationship between *M. genitalium* and human papillomavirus (HPV) is an emerging area of investigation with oncological implications that merit careful framing. Co-infection prevalence figures vary by population and setting: in a 2024–2025 Spanish prospective cohort of women followed for HR-HPV infection, *M. genitalium* and other STIs were detected at marginally higher rates in HPV-positive (5.1%) than HPV-negative women (3.8%).37 In a large Tanzanian cohort, *M. genitalium* was significantly more prevalent among women with HPV, and mycoplasma infections — particularly *M. hominis* — were associated with higher-grade cervical lesions.38 A 2018 meta-analysis found *M. genitalium* associated with a significantly increased risk of high-risk HPV infection (OR 1.50).39 A 2025 systematic review and meta-analysis of six case-control studies found genital mycoplasmas associated with a significantly increased risk of cervical cancer overall (OR 1.64; 95% CI 1.25–2.14).40

The proposed biological mechanism is plausible but not yet definitively established: *M. genitalium* infection reduces microvilli in cervical epithelial cells, releases pro-inflammatory cytokines, and disrupts the physical and local immune barriers of the cervix, potentially creating favorable conditions for HPV penetration of basal cells and facilitating persistent HPV infection — the prerequisite for cervical carcinogenesis.39 The chronic NF-κB activation driven by *M. genitalium* membrane lipoproteins parallels the inflammatory carcinogenesis pathway most thoroughly characterized in *Helicobacter pylori*-associated gastric cancer, where decades of NF-κB-driven mucosal inflammation produce a progression from gastritis to intestinal metaplasia to adenocarcinoma. Whether an analogous long-term progression occurs in *M. genitalium*-infected urogenital epithelium remains uncharacterized. Importantly, the evidence is not uniformly consistent: a Chinese colposcopy cohort

found no significant association between *M. genitalium* co-infection and the severity of cervical lesions in HPV-positive women,41 and a retrospective Sichuan cohort found no association with high-risk HPV infection.42 The oncological hypothesis is therefore biologically plausible, epidemiologically supported in some populations, and mechanistically grounded — but not yet established with the rigor required for clinical guideline incorporation. Prospective studies controlling for shared sexual risk behaviors, HPV genotype, and vaginal microbiome composition are needed before causal conclusions can be drawn.

3.11.5 Implications for the Present Case

The patient described in this report had microbiologically confirmed active *M. genitalium* infection from early October 2025 through February 27, 2026 — approximately five months of documented persistent infection. During this period he experienced continuous urethral inflammation (urinalysis demonstrating 26–50 WBCs/hpf), repeated antibiotic-driven inflammatory cycling adding iatrogenic mucosal stress at each treatment course, two episodes of leukopenia consistent with systemic hematologic impact of prolonged antimicrobial exposure, and at minimum one episode of antibiotic chelation producing subtherapeutic drug levels. Although cystourethroscopy performed during the active infection period demonstrated no stricture or scarring, the cumulative inflammatory burden over five months of active infection with five antibiotic courses is non-trivial. Long-term urological follow-up for stricture surveillance is warranted. The impact on the female partner's reproductive tract health, and the adequacy of her treatment during the prolonged window of male partner infectivity, remains to be determined.

3.12 Research Priorities

This case generates five specific, practical research questions that are timely, feasible with existing infrastructure, and directly relevant to improving outcomes for future patients.

3.12.1 US Resistance Surveillance Cohort

The most urgent unmet need in US *M. genitalium* management is a systematic, nationally representative resistance prevalence dataset. Current US macrolide resistance estimates (≥64%) are extrapolated from small convenience samples and selected clinic populations rather than from structured surveillance. A prospective, multicenter NAAT-based prevalence and resistance genotyping study in US STI clinics and primary care settings — performing reflex 23S rRNA and parC/gyrA genotyping on all NAAT-positive specimens — would provide the epidemiological

foundation needed to calibrate treatment guidelines and track resistance trends over time. A single-institution pilot study in a rural, non-urban population such as the Southern Tier of New York — substantially underrepresented in existing cohorts — would be independently publishable and could serve as a template for multicenter expansion.

3.12.2 Pharmacokinetic Failure Registry

The chelation-mediated treatment failure documented in this case is almost certainly not unique to this patient — it is unique in being recognized and documented. A prospective registry systematically eliciting supplement and mineral co-administration in all patients prescribed moxifloxacin for *M. genitalium* — capturing supplement name, dose, iron/zinc/calcium/magnesium content, and timing relative to antibiotic dosing — linked to TOC-confirmed cure or failure, would allow the incidence of pharmacokinetic treatment failure attributable to chelation to be estimated prospectively. This study requires no special laboratory infrastructure: a structured intake questionnaire at the time of prescription and TOC result linkage at ≥3 weeks are sufficient. The data generated would directly quantify a preventable cause of treatment failure and provide the evidence base for standardized counseling interventions at all moxifloxacin prescribing encounters.

3.12.3 Supplement Counseling Intervention Trial

A before-and-after quality improvement study implementing a standardized supplement avoidance counseling protocol at the time of moxifloxacin prescription — an explicit written instruction set identifying chelating agents, minimum separation intervals, and the clinical consequences of co-administration — with TOC-confirmed cure rates as the primary outcome, is the most immediately actionable study arising from this case. It requires no external funding at baseline, no new laboratory capability, and no patient population beyond those already receiving standard-of-care treatment. If a structured counseling intervention demonstrably improves TOC-confirmed cure rates, the result is publishable as a quality improvement study and has direct implementation value at any prescribing institution.

3.12.4 Silent PID Prevalence in Female Partners

The evidence base linking *M. genitalium* to upper tract reproductive disease in women is largely retrospective. A prospective endometrial biopsy study in female partners of men with confirmed *M. genitalium* urethritis — sampling at the time of partner notification and treatment, and again following confirmed cure — would directly quantify the prevalence of subclinical endometritis in this

population and establish whether early treatment prevents upper tract progression. This is the foundational evidence type that drove the USPSTF chlamydia screening recommendation. *M. genitalium* requires equivalent prospective data to support a comparable public health response.

3.12.5 Anti-Adhesin and Pro-Desquamation Validation in Cell Culture

The therapeutic framework developed in Section 3.10 — targeting the P140/P110 tip organelle adhesin complex and accelerating epithelial desquamation to deprive the organism of its colonization substrate — is currently hypothetical. In vitro validation is the appropriate immediate next step. Specifically: whether N-acetylcysteine, heparin-like glycosaminoglycans, or lactoferrin at clinically achievable concentrations reduce *M. genitalium* adherence to urethral epithelial cell lines; whether combining adhesin blockade with doxycycline produces additive or synergistic reduction in viable organism count compared to either agent alone; and whether mechanical or chemical desquamation of *M. genitalium*-infected epithelial cell monolayers reduces recovered organism burden in a dose-dependent fashion. The intellectual framework for this work has been laid in the present report and could serve as the conceptual foundation for a grant application or academic laboratory collaboration. Critically, because this approach targets host-pathogen interaction rather than bacterial metabolism, it is entirely resistance-agnostic — its efficacy would be unaffected by macrolide resistance, fluoroquinolone resistance, or the emergence of resistance to future antibiotic classes.

4. Conclusions

This case demonstrates three independent and compounding barriers to cure in *M. genitalium* urethritis, each representing a translatable clinical lesson:

- Diagnostic delay and empiric azithromycin without resistance testing: In high-resistance settings (US macrolide resistance ≥64%), initiating azithromycin without prior resistance genotyping is effectively driving resistance selection in the majority of treated patients. Clinicians in the US should advocate for laboratory-developed resistance testing platforms and request them when available.
- The unique genomic architecture of M. genitalium (single-copy 23S rRNA gene, absent MMR) makes it biologically prone to rapid, irreversible resistance acquisition from even a single course of inadequate therapy.
- Fluoroquinolone-mineral chelation from dietary supplements is a pharmacokinetically significant, preventable cause of treatment failure that requires systematic elicitation as part of medication history-taking in all patients with M. gen treatment failure.
- Prolonged antibiotic courses require CBC monitoring; leukopenia and neutropenia are real adverse effects of extended doxycycline-fluoroquinolone combinations.

Antimicrobial stewardship for *M. genitalium* is at a critical inflection point. Resistance rates approaching those seen with gonorrhea, combined with the absence of robust salvage options in the United States, underscore the urgency of resistance-guided approaches, PK-aware prescribing, and rigorous pharmacological counseling at each treatment encounter.

5. Key Learning Points for Clinicians

Domain	Clinical Take-Away
Diagnosis	Order NAAT for M. gen in all persistent/recurrent NGU. Request resistance genotyping if available; document susceptibility before selecting second-line agent.
First-line treatment	Doxycycline 100 mg BID × 7d as initial step-down, then resistance-guided second agent. Do not use azithromycin 1g empirically in high-resistance settings.
Fluoroquinolone counseling	Explicitly instruct patients to avoid all multivitamins, mineral supplements, dairy, antacids, and iron-containing products during moxifloxacin dosing and for 4–6 hours before/after each dose.
Treatment failure workup	Elicit full medication and supplement history. Consider PK failure before assuming microbiological resistance as the sole cause.
Test of cure	Perform TOC ≥3 weeks after treatment completion. Earlier negative NAATs may reflect residual non-viable DNA, not cure.
Hematologic monitoring	Obtain CBC at baseline and every 2–4 weeks during antibiotic courses ≥14 days, especially with fluoroquinolone-tetracycline combinations.
Natural history	Untreated M. gen infection persists a mean of ~15 months. Prolonged persistence increases transmission risk and resistance selection window.

6. Disclosures

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References

1. Bachmann LH, Kirkcaldy RD, Geisler WM, et al. Prevalence of *Mycoplasma genitalium* infection, antimicrobial resistance mutations and symptom resolution following treatment of urethritis. *Clin Infect Dis.* 2020;71:e624–e632.

2. Taylor-Robinson D, Jensen JS. *Mycoplasma genitalium*: from chrysalis to multicolored butterfly. *Clin Microbiol Rev.* 2011;24:498–514.

3. Machalek DA, Tao Y, Shilling H, et al. Prevalence of mutations associated with resistance to macrolides and fluoroquinolones in *Mycoplasma genitalium*: a systematic review and meta-analysis. *Lancet Infect Dis.* 2020;20:1302–1314.

4. Workowski KA, Bachmann LH, Chan PA, et al. Sexually transmitted infections treatment guidelines, 2021. *MMWR Recomm Rep.* 2021;70(4):1–187.

5. Jensen JS, Cusini M, Gomberg M, et al. 2021 European guideline on the management of *Mycoplasma genitalium* infections. *J Eur Acad Dermatol Venereol.* 2022;36:641–650.

6. Alliance ASH. Australian STI management guidelines: *Mycoplasma genitalium.* 2020. http://www.sti.guidelines.org.au

7. Pitman SK, Frei CR. Fluoroquinolone-cation interactions: a systematic review. *Pharmacy.* 2019;7(3):104.

8. Baumann L, Cina M, Egli-Gany D, et al. Prevalence of *Mycoplasma genitalium* in different population groups: systematic review and meta-analysis. *Sex Transm Infect.* 2018;94:254–261.

9. Smieszek T, White PJ. Apparently-different clearance rates from cohort studies of *Mycoplasma genitalium* are consistent after accounting for incidence of infection, recurrent infection, and study design. *PLoS ONE.* 2016;11(2):e0149087.

10. Fraser CM, Gocayne JD, White O, et al. The minimal gene complement of *Mycoplasma genitalium. Science.* 1995;270:397–403.

11. Manhart LE, Geisler WM, Bradshaw CS, Jensen JS, Martin DH. Weighing potential benefits and harms of *Mycoplasma genitalium* testing and treatment approaches. *Emerg Infect Dis.* 2022;28(8):e1–e11.

12. Murray GL, Bodiyabadu K, Danielewski J, et al. Moxifloxacin and sitafloxacin treatment failure in *Mycoplasma genitalium* infection: association with parC mutation G248T (S83I) and concurrent gyrA mutations. *J Infect Dis.* 2020;221:1017–1024.

13. Stass H, Kubitza D. Effects of iron supplements on the oral bioavailability of moxifloxacin, a novel 8-methoxyfluoroquinolone, in humans. *Clin Pharmacokinet.* 2001;40(Suppl 1):57–62. PMID: 11352443.

14. Declercq S, et al. M3P combination therapy for multidrug-resistant *Mycoplasma genitalium*: a case series. *Sex Transm Infect.* 2026. PMID: 40335273.

15. Luria SE, Delbruck M. Mutations of bacteria from virus sensitivity to virus resistance. *Genetics.* 1943;28(6):491–511.

16. Molloy SF, Kanyama C, Heyderman RS, et al. Antifungal combinations for treatment of cryptococcal meningitis in Africa (ACTA). *N Engl J Med.* 2018;378:1004–1017.

17. Malfertheiner P, Megraud F, Rokkas T, et al. Management of *Helicobacter pylori* infection: the Maastricht VI/Florence consensus report. *Gut.* 2022;71(9):1724–1762.

18. Liu C, Bayer A, Cosgrove SE, et al. Clinical practice guidelines by the IDSA for the treatment of methicillin-resistant *Staphylococcus aureus* infections in adults and children. *Clin Infect Dis.* 2011;52(3):e18–e55.

19. Glass JI, Assad-Garcia N, Alperovich N, et al. Essential genes of a minimal bacterium. *Proc Natl Acad Sci USA.* 2006;103(2):425–430.

20. Hutchison CA, Chuang RY, Noskov VN, et al. Design and synthesis of a minimal bacterial genome. *Science.* 2016;351(6280):aad6253.

21. Jensen JS, Bradshaw CS, Tabrizi SN, Fairley CK, Hamasuna R. Azithromycin treatment failure in *Mycoplasma genitalium*-positive patients with nongonococcal urethritis is associated with induced macrolide resistance. *Clin Infect Dis.* 2008;47:1546–1553.

22. Rottem S. Interaction of mycoplasmas with host cells. *Physiol Rev.* 2003;83(2):417–432.

23. Burgos R, Pich OQ, Ferrer-Navarro M, Baseman JB, Querol E, Pinol J. *Mycoplasma genitalium* P140 and P110 cytadhesins are reciprocally stabilized and required for cell adhesion and terminal-organelle development. *J Bacteriol.* 2006;188(24):8627–8637.

24. Mulvey MA, Schilling JD, Hultgren SJ. Establishment of a persistent *Escherichia coli* reservoir during the acute phase of a bladder infection. *Infect Immun.* 2001;69(7):4572–4579.

25. Kannan TR, Provenzano D, Wright JR, Baseman JB. Identification and characterization of human surfactant protein A binding protein of *Mycoplasma pneumoniae*. Infect Immun. 2005;73(4):2828–2834. [CARDS toxin structural characterization]

26. Vielot NA, Smith JS, Hudgens MG, et al. *Mycoplasma genitalium* and cytokine and chemokine responses in the female genital tract. *Sex Transm Dis.* 2018;45(12):788–793.

27. Horner PJ. Mycoplasma genitalium: an emerging cause of sexually transmitted disease in men. *Sex Transm Infect.* 2012;88(7):508–510.

28. Svenstrup HF, Fedder J, Kristoffersen SE, Trolle B, Birkelund S, Christiansen G. *Mycoplasma genitalium* attaches to human spermatozoa. *Hum Reprod.* 2003;18(10):2103–2109.

29. Lis R, Rowhani-Rahbar A, Manhart LE. *Mycoplasma genitalium* infection and female reproductive tract disease: a meta-analysis. *Clin Infect Dis.* 2015;61(3):418–426.

30. Haggerty CL, Totten PA, Tang G, et al. Identification of novel microorganisms associated with pelvic inflammatory disease. *Sex Transm Infect.* 2016;92(6):441–446.

31. Bhugra B, Bhugra B, Voelker LL, Bhugra B, Dybvig K. Mechanism of antigenic variation in *Mycoplasma pulmonis.* Proc Natl Acad Sci USA. 1995;92(25):11@754–11758.

32. Ueno PM, Timenetsky J, Centonze VE, et al. Interaction of *Mycoplasma genitalium* with host cells: evidence for nuclear localisation. *Microbiology.* 2008;154(Pt 12):3459–3469.

33. Mavedzenge SN, Van Der Pol B, Weiss HA, et al. The association between *Mycoplasma genitalium* and HIV-1 acquisition in African women. *J Infect Dis.* 2012;206(11):1718–1726.

34. Taylor SN, Marrazzo J, Batteiger BE, et al. *Mycoplasma genitalium* prevalence, coinfection, and macrolide antibiotic resistance frequency in a multicenter clinical study cohort in the United States. *J Clin Microbiol.* 2019;57(12):e01053-16.

35. Lillis RA, Nsuami MJ, Myers L, Martin DH. Utility of urine, vaginal, cervical, and rectal specimens for detection of *Mycoplasma genitalium* in women attending a sexually transmitted diseases clinic. *J Clin Microbiol.* 2011;49(5):1990–1992.

36. Khosropour CM, Jensen JS, Soge OO, et al. *Mycoplasma genitalium* co-infection in women with *Chlamydia trachomatis* infection. *Sex Transm Dis.* 2020;47(12):840–844.

37. Aznarte VM, Marco-Fuertes A, Ortega-Martínez M, et al. HPV-associated sexually transmitted infections in cervical cancer screening: a prospective cohort study. *Viruses.* 2025;17(2):247.

38. Baker J, Wei S, Wood C, et al. Mycoplasma co-infection is associated with cervical cancer risk. *Cancers (Basel).* 2020;12(5):1093.

39. Ye G, Tang W, Peng R, et al. Association of *Mycoplasma genitalium* infection with female reproductive system diseases: a meta-analysis. *Eur J Clin Microbiol Infect Dis.* 2018;37(3):421–432.

40. Obeagu EI, Obeagu GU, Omo-Emmanuel UK, et al. Genital mycoplasma infections: a hidden factor in cervical cancer progression? A systematic review and meta-analysis. *BMC Infect Dis.* 2025;25(1):1309.

41. Jiang M, Ding H, He L, et al. Association between co-infection with *Chlamydia trachomatis* or *Mycoplasma genitalium* and cervical lesions in HPV-positive population in Hunan, China: a cross-sectional study. *Infect Agents Cancer.* 2023;18(1):76.

42. Wang X, Shen W, Zhang Y, et al. *Mycoplasma genitalium* and *Chlamydia trachomatis* infection among women in Southwest China: a retrospective study. *BMC Womens Health.* 2022;22(1):291.